

Project Report

Demand Forecasting & Inventory Optimization

This report is generated from the repository's deterministic synthetic sample. It demonstrates method execution only; results are not business evidence.

Scope and workflow

The sample contains 8 SKUs and 540 daily dates. Transactions are cleaned, net returns into SKU-day demand, and are evaluated with a chronological holdout. The selected forecast is translated into service-level stock policy estimates.

Illustrative holdout comparison

Model	MAE (units)	RMSE (units)	WAPE
hist_gradient_boosting	3.56	4.66	24.6%
seasonal_naive	4.43	5.85	30.6%
moving_average_7	4.44	5.62	30.7%
naive	4.69	6.48	32.4%

Best WAPE on this synthetic holdout: hist_gradient_boosting (24.6%). Suggested replenishment review count: 2 of 8 SKUs. Neither value should be interpreted as an expected result on other data.

Decision formulas

Safety stock = $Z \times \text{daily demand standard deviation} \times \sqrt{\text{fixed lead time}}$. Reorder point = mean daily demand $\times$ lead time + safety stock. EOQ = $\sqrt{2 \times \text{annual demand} \times \text{order cost} / \text{annual holding cost per unit}}$. These are simplified assumptions and omit purchase orders, uncertain lead time, capacity, expiry, and supplier calendars.

Recommended interpretation

Use this project to demonstrate reproducible engineering, time-series validation, and explainable policies. Do not claim inventory savings, recovered demand, or guaranteed service levels without a validated operational pilot.